

Adaptive Diversity-Enhanced Adversarial Reinforcement Learning for Robust Autonomous Driving

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ABSTRACT

Robust policy learning under uncertainty remains challenging for autonomous driving agents trained with reinforcement learning, particularly due to optimization instability, redundant continuous actions, and adversarial policy collapse. Existing diversity regularization methods mainly rely on point-wise or pairwise divergence measures, which are insufficient to characterize the global geometric structure of multi-policy action distributions. To address these limitations, this paper proposes an adaptive diversity-enhanced adversarial reinforcement learning framework. {The key idea of this work is to measure the diversity of a set of adversarial policies geometrically: their responses to a shared batch of states are arranged as a joint action matrix, and the log-volume of its singular value spectrum is used as a differentiable measure of how large a subspace the set spans, with the strength of this objective regulated online by feedback from the learning process.} In addition, a multi-source disturbance generation strategy exposes the protagonist policy to {sensor noise and vehicle dynamics variations}. Experiments in a high-fidelity autonomous driving simulation benchmark under {sensor noise and randomized vehicle dynamics} demonstrate that the proposed framework achieves better safety, robustness, and cross-condition generalization than representative baselines. {Across six cross-regime settings, our algorithm improves average returns by 7.2% over the strongest baseline and 30.0% over PPO, yielding 71%–91% safe-completion rates (vs. baselines' 13%–90%). Ablations verify each component's necessity: removing spectral diversity, adaptive regulation, multi-head architecture, and multi-source perturbation reduces returns by up to 21%, 25%, 34%, and 67%, respectively.